

## Problem

Two coils are mutually coupled, with  $L_1 = 50 \text{ mH}$ ,  $L_2 = 120 \text{ mH}$ , and  $k = 0.5$ . Calculate the maximum possible equivalent inductance if:

(a) the two coils are connected in series

(b) the coils are connected in parallel

## Step-by-step solution

## Step 1 of 5

Consider the following data for the for two mutually coupled coils.

$$L_1 = 50 \text{ mH}$$

$$L_2 = 120 \text{ mH}$$

Coefficient of coupling,  $k = 0.5$

Calculate the mutual inductance.

$$\begin{aligned} M &= k\sqrt{L_1 L_2} \\ &= 0.5\sqrt{50 \times 10^{-3} \times 120 \times 10^{-3}} \\ &= 38.72 \text{ mH} \end{aligned}$$

## Step 2 of 5

(a)

If two coils are connected in series maximum possible inductance will get only when the two coils are connected in series aiding.

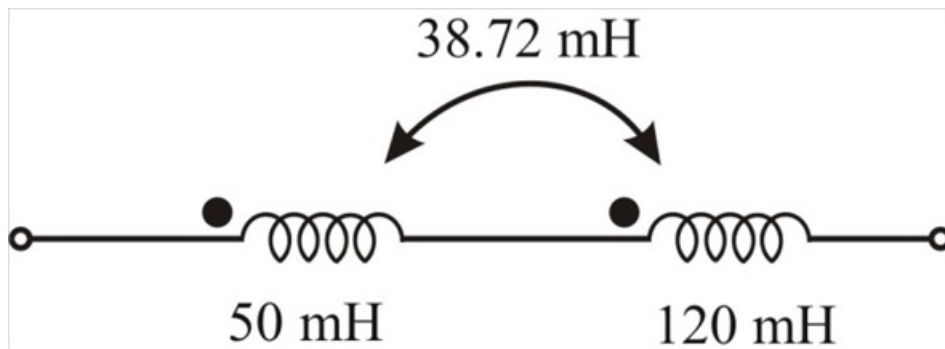


Figure 1

## Step 3 of 5

Calculate the equivalent inductance.

$$\begin{aligned}L_{eq} &= L_1 + L_2 + 2M \\&= (50 + 120 + 38.72) \times 10^{-3} \\&= 247.45 \text{ mH}\end{aligned}$$

Maximum possible inductance when coils are connected in series is 247.45 mH.

[Comments \(1\)](#)



**Anonymous**

THIQ BITCHES ONLY but real talk they got the right answer but they forgot to multiply by 2.

**Step 4 of 5**

(b)

If two coils are connected in parallel, maximum possible inductance will get only when the two coils are connected in parallel aiding.

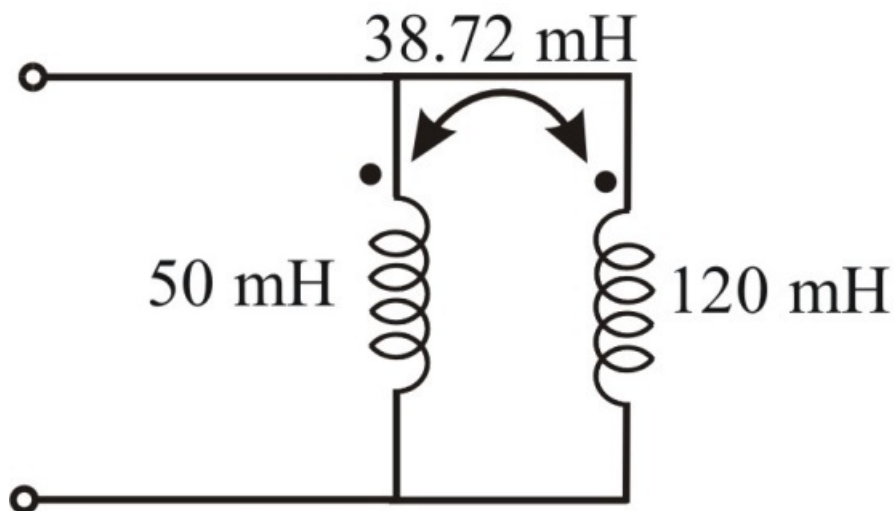


Figure 2

**Step 5 of 5**

Calculate the equivalent inductance.

$$\begin{aligned}L_{eq} &= \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \\&= \frac{(50 \times 10^{-3})(120 \times 10^{-3}) - (38.72 \times 10^{-3})^2}{50 \times 10^{-3} + 120 \times 10^{-3} - 2 \times 38.72 \times 10^{-3}} \\&= 48.62 \text{ mH}\end{aligned}$$

Maximum possible inductance when coils are connected in parallel is 48.62 mH.